JIARUI WANG

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Summary

I am a Ph.D. student at the School of Computer Science, Shanghai Jiao Tong University. Currently, I am working on LLM pretraining, including data pipelines, model training, and scaling experiments. I also led the research and development of *Intern-S2-Preview*'s memory architecture for continual expansion into frontier scientific domains.

Education

Shanghai Jiao Tong University, Shanghai, China — Sept. 2025 – Present
Ph.D. in Computer Science and Technology
Advisor: Prof. **Zhouhan Lin** (LUMIA Group)

Shanghai Jiao Tong University, Shanghai, China — Sept. 2021 – Jun. 2025
B.S. in Computer Science and Technology

Work Experience

LLM Research Intern, Shanghai AI Lab — Aug. 2025 – Present

- **LLM Pretraining and Midtraining:** Built multilingual data pipelines to process raw PDFs and web pages, filter data across languages, and predict quality, topic, and other metadata labels for pretraining and midtraining corpora. Conducted scaling experiments to evaluate model architectures and data mixtures, select training configurations, and improve performance at larger model scales.
- **Model Architecture:** Led research and development of *Intern-S2-Preview*'s memory architecture, decoupling domain specialization from backbone parameter updates. The architecture enables continual expansion into frontier scientific domains through independently trained, plug-and-play memories.
- **Scientific Domain SFT:** Conducted supervised fine-tuning of *Intern-S2-Preview*'s memories for multiple frontier scientific domains, covering data preparation, training, and evaluation.
- **Infrastructure:** Designed and implemented a large-scale GPU-parallel retrieval system and extended the inference engine to support memory-augmented decoding, enabling scalable retrieval and inference for new architectures.

Publications

- **MemSFT: Mitigating Alignment Tax with an External Parametric Memory**
  Jiarui Wang[*], Xiang Shi[*], Jiaqi Cao, Rubin Wei, Xiquan Wang, Hao Sun, Jingzhi Wang, Zhiqi Yang, Qipeng Guo, Bowen Zhou, Zhouhan Lin
  arXiv:2607.25614, 2026.
- **Intern-S2-Preview: Scientific Agentic Foundation Model**
  Intern-S2-Preview Team, Shanghai AI Laboratory; Co-author; *arXiv:2608.13505, 2026.*
- **Memory Decoder at Scale: A Pretrained, Parametric Long-Term Memory**
  Rubin Wei, Jiaqi Cao, **Jiarui Wang**, Junming Zhang, Qipeng Guo, Bowen Zhou, Zhouhan Lin
  arXiv:2607.27919, 2026.
- **MLP Memory: A Retriever-Pretrained Memory for Large Language Models**
  Rubin Wei[*], Jiaqi Cao[*], **Jiarui Wang**, Jushi Kai, Qipeng Guo, Bowen Zhou, Zhouhan Lin
  ICLR 2026.
- **Memory Decoder: A Pretrained, Plug-and-Play Memory for Large Language Models**
  Jiaqi Cao[*], **Jiarui Wang**[*], Rubin Wei, Qipeng Guo, Kai Chen, Bowen Zhou, Zhouhan Lin
  NeurIPS 2025.

[*]Equal contribution.

Honors and Awards

Second Place, RoboCup China Open, Small Size League, 2024
Third Prize, Prototype System Competition, CCF ChinaSoft, 2023
First Prize (Provincial), Senior Division, National Olympiad in Informatics in Provinces (NOIP), 2018